

Module 2: Network + Network Security , Maintenance and Troubleshooting

Networking Troubleshooting, Management , Monitoring, Optimization

Task 1:

```
C:\Users\aasha>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Aashanap
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : localdomain


Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : localdomain
Description . . . . . : vmxnet3 Ethernet Adapter
Physical Address. . . . . : 00-0C-29-F5-35-A7
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::24a6:ba6:bdc8:973c%14(Preferred)
IPv4 Address. . . . . : 192.168.249.131(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 10 August 2026 13:54:11
Lease Expires . . . . . : 10 August 2026 16:03:25
Default Gateway . . . . . : 192.168.249.2
DHCP Server . . . . . : 192.168.249.254
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-C3-AC-79-00-0C-29-F5-35-A7
DNS Servers . . . . . : 192.168.249.2
Primary WINS Server . . . . . : 192.168.249.2
NetBIOS over Tcpip. . . . . : Enabled
```

IP - 192.168.249.131

Subnet Mask - 255.255.255.0

Default Gateway - 192.168.249.2

DNS servers - 192.168.249.2

Task 2:

With Connection

```
aashanapatel@192 ~ % ping www.google.com
PING www.google.com (142.251.150.119): 56 data bytes
64 bytes from 142.251.150.119: icmp_seq=0 ttl=118 time=28.407 ms
64 bytes from 142.251.150.119: icmp_seq=1 ttl=118 time=20.889 ms
64 bytes from 142.251.150.119: icmp_seq=2 ttl=118 time=15.444 ms
64 bytes from 142.251.150.119: icmp_seq=3 ttl=118 time=14.438 ms
64 bytes from 142.251.150.119: icmp_seq=4 ttl=118 time=15.167 ms
64 bytes from 142.251.150.119: icmp_seq=5 ttl=118 time=14.691 ms
64 bytes from 142.251.150.119: icmp_seq=6 ttl=118 time=14.955 ms
64 bytes from 142.251.150.119: icmp_seq=7 ttl=118 time=15.150 ms
64 bytes from 142.251.150.119: icmp_seq=8 ttl=118 time=14.921 ms
```

Without Connection

```
aashanapatel@192 ~ % ping www.google.com
ping: cannot resolve www.google.com: Unknown host
```

Task 3:

Monitor Network Usage While Streaming YouTube

App Name	Energy Impact	12 hr Po...	App Nap	Preventing Sl...
> VMware Fusion	1.4	56.69	No	No
> Pages	0.1	50.63	No	No
▼ Safari	13.7	49.44	No	Yes
com.apple.Safari.SearchHelper	0.1	-	No	No
https://www.youtube.com	175.2	-	No	Yes

Task 4:

No.	Time	Source	Destination	Protocol	Length	Info
36	1.072051	17.242.13.5	192.168.1.5	TLSv1...	421	Application Data
38	1.084617	192.168.1.5	17.242.13.5	TLSv1...	98	Application Data
84	2.453911	192.168.1.5	49.44.140.153	TLSv1...	430	Client Hello (SNI=www.bing.com)
86	2.481998	49.44.140.153	192.168.1.5	TLSv1...	1494	Server Hello, Change Cipher Spec, Application Data
91	2.483921	17.242.13.5	192.168.1.5	TLSv1...	421	Application Data
95	2.497757	192.168.1.5	17.242.13.5	TLSv1...	98	Application Data
97	2.505319	49.44.140.153	192.168.1.5	TLSv1...	1000	Application Data, Application Data, Application Data
99	2.506189	192.168.1.5	49.44.140.153	TLSv1...	96	Change Cipher Spec, Application Data
118	2.722507	192.168.1.5	17.253.18.195	TLSv1...	583	Client Hello (SNI=configuration.apple.com)
120	2.736976	17.253.18.195	192.168.1.5	TLSv1...	1506	Server Hello, Change Cipher Spec
122	2.736978	17.253.18.195	192.168.1.5	TLSv1...	1417	Application Data
124	2.744804	192.168.1.5	17.253.18.195	TLSv1...	130	Change Cipher Spec, Application Data
125	2.745355	192.168.1.5	17.253.18.195	TLSv1...	494	Application Data
127	2.763318	17.253.18.195	192.168.1.5	TLSv1...	968	Application Data, Application Data

3 types of information visible even in encrypted (HTTPS/TLS) traffic:

1. **Server hostname (SNI)** — in the Client Hello packet, the Server Name Indication field reveals which domain you're connecting to (e.g., www.flipkart.com), even though the actual page content stays encrypted.
2. **Source and destination IP addresses and ports** — you can see your machine's IP talking to the server's IP on port 443, revealing *who* you're communicating with, just not *what* is being said.
3. **Certificate details** — the Server Hello/Certificate exchange shows the site's SSL/TLS certificate info (issuer, validity dates, domain it's issued for), which you can inspect even without decrypting the session.

Task 5:

AI-Generated Troubleshooting Checklist for Slow Gaming Internet

Here's a checklist you can paste into your assignment:

1. **Restart your router and modem** — power cycle both for 30 seconds; this clears temporary memory leaks and reconnects your ISP session, often resolving speed drops instantly.
2. **Switch from Wi-Fi to a wired Ethernet connection** — Wi-Fi introduces latency variability (jitter) that's especially disruptive for real-time gaming; Ethernet gives a stable, low-latency path.
3. **Check for bandwidth-hogging devices/apps** — other devices on the network doing large downloads, cloud backups, or 4K streaming can starve your game's connection; pause them during play.
4. **Update your router's firmware and place it centrally** — outdated firmware can cause performance bugs, and physical obstructions/distance weaken Wi-Fi signal strength.
5. **Enable Quality of Service (QoS) settings on your router** — QoS lets you prioritize gaming traffic over other household traffic, reducing lag spikes when others are using the network.
6. **Contact your ISP if speeds are consistently below your plan** — run a speed test at different times of day; if you're consistently far below your subscribed speed, there may be a line issue or local congestion your ISP needs to address.

